



Customer Churn Early-Warning Analysis

A complete SQL risk-scoring and Power BI portfolio case study

VALIDATED OUTCOME

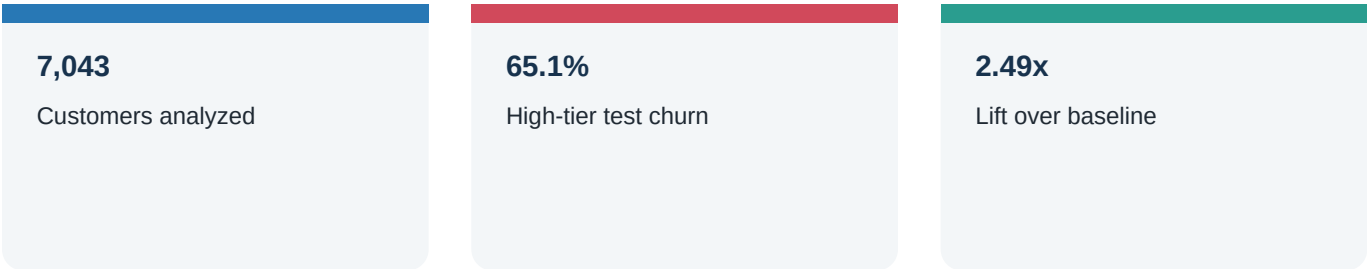
High-risk customers recorded 65.1% churn on an untouched test sample, compared with 26.1% overall. The score delivered 2.49x lift and captured 50.3% of churners within 20.2% of customers.

Technology

SQL | SQLite | Power BI | DAX | Data storytelling

Executive Summary

The project in one page



- Business problem: retention teams need a prioritized list before customers leave.
- Solution: an interpretable 0-100 score built from eight observable account signals.
- Validation: signal weights learned on 80%; final performance measured on a held-out 20%.
- Action: contact High-risk customers first and tailor outreach using each customer's dominant reason.

Tier	Customers	Churners	Churn rate
High	284	185	65.1%
Medium	442	134	30.3%
Low	683	49	7.2%

Business Problem

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Churn is costly because the organization often recognizes it only after a customer has already left.

Decision to support

Which current customers should the retention team contact first, why were they selected, and what intervention is appropriate? The project turns historical churn patterns into a ranked operational queue.

Business users

- Retention managers need prioritization, not another static churn-rate chart.
- Customer-success agents need a plain-language reason for every flag.
- Executives need evidence that the proposed tier separates churners from non-churners.

Success definition

A useful system should concentrate actual churners in a manageable High-risk group, remain explainable, avoid target leakage, and create outputs that can be refreshed and loaded into Power BI.

Objectives and Scope

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The project is a transparent decision-support system, not a claim of perfect prediction.

Primary objectives

- Create a clean customer-level analytical model in SQL.
- Derive weights from observed churn behavior instead of guessing them.
- Assign a 0-100 score, risk tier, and top reason to every customer.
- Validate performance on customers not used to construct the weights.
- Translate the result into retention actions and dashboard requirements.

Out of scope

The dataset does not contain event dates, marketing costs, support tickets, complaints, failed payments, location, or campaign outcomes. Therefore the project does not estimate causal financial impact or claim recent behavior changes.

Ethical scope

Demographic fields can be used to audit outcomes, but should not determine whether a customer receives beneficial retention treatment.

Dataset Overview

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The IBM Telco Customer Churn sample provides a realistic customer snapshot with 21 original columns.

Population

7,043 telecom customers. Each row represents one customer and the Churn field records Yes or No. The observed overall dataset churn rate is approximately 26.5%.

Field groups

- Customer context: gender, senior status, partner, dependents.
- Relationship: tenure, contract, phone service, internet service.
- Service adoption: security, backup, protection, technical support, streaming.
- Billing: paperless billing, payment method, monthly charges, total charges.
- Outcome: churn label.

Why it fits

The data is small enough for a fully reproducible SQLite project but rich enough to demonstrate modeling judgment, validation, SQL window functions, business interpretation, and dashboard design.

Data Quality and Preparation

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Data checks protect the score from silent errors and make the workflow reproducible.

Checks completed

- 7,043 rows loaded successfully.
- Zero duplicated customer IDs.
- Zero invalid churn values.
- Eleven blank TotalCharges values identified; these correspond to customers with effectively no billing history and are stored as NULL.

Transformations

Column names were standardized, numeric types were applied to tenure and charges, Churn was converted into a binary churned flag, and descriptive tenure and monthly-charge bands were created.

Reproducibility

The build script recreates the SQLite database from the raw CSV, executes schema and modeling SQL, exports four dashboard-ready CSV files, and prints validation results in one command.

SQL Architecture

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A layered SQL design separates ingestion, reusable features, scoring, and reporting.

Layer 1 - Raw table

`raw_customers` preserves one typed record per source customer. The customer ID is the primary key.

Layer 2 - Feature view

`customer_features` adds the churn flag, deterministic train/test sample, tenure band, charge band, and eight binary risk signals.

Layer 3 - Analytical tables

`signal_weights` stores training-only churn lift. `customer_scores` stores normalized risk, tier, and dominant reason. Long-form signals simplify weighting and ranking.

Layer 4 - Reporting views

`customer_risk_export` supplies the action list; `validation_by_tier` supplies held-out performance. CSV exports feed Power BI without complex relationships.

Feature Engineering

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Eight signals were selected because they are available before the outcome and understandable to business users.

Signals

- Month-to-month contract.
- Tenure of 12 months or less.
- Fiber optic internet service.
- No online security.
- No technical support.
- Electronic-check payment.
- Paperless billing.
- Monthly charge of \$70 or more.

Design principle

Every signal is binary and traceable to a source attribute. This makes the calculation auditable: an analyst can explain exactly which conditions increased a customer score.

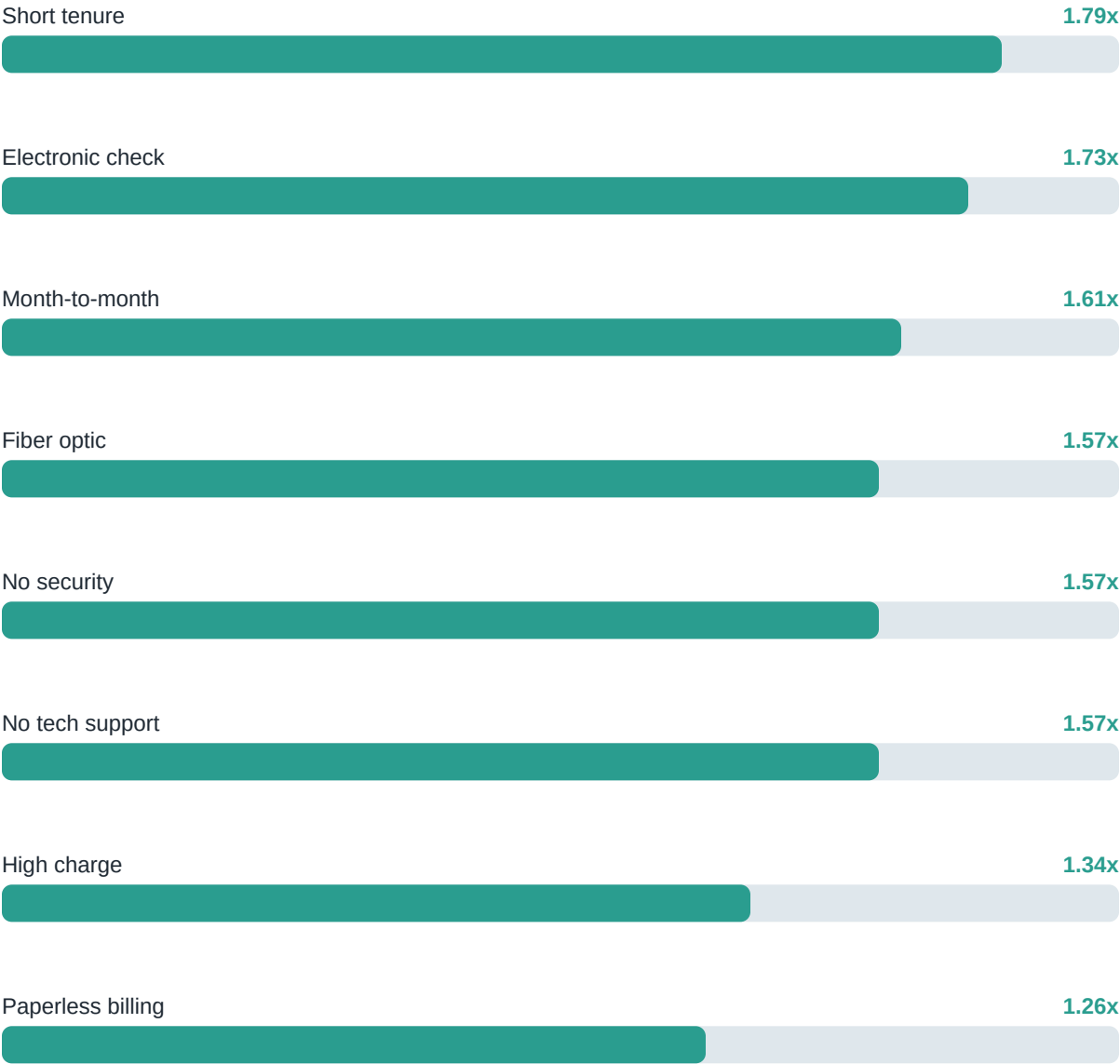
What was not created

No usage decline, recent ticket count, billing failure, or 30-day trend was invented. A snapshot cannot support those time-based claims.

Signal Strength

Observed training-sample churn lift

Weights are derived only from the 80% training population.



Interpretation: short-tenure customers churned at 1.79 times the training baseline. Lift indicates association, not causation.

Risk-Score Methodology

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The calculation combines positive churn lift and rescales the result for prioritization.

Step 1 - Baseline

Calculate the average churn rate across the training sample.

Step 2 - Signal weight

For each signal: $\text{weight} = \max(0, \text{signal churn rate} / \text{baseline churn rate} - 1)$. Negative lift is floored at zero because the score represents elevated risk.

Step 3 - Customer total

For each customer, multiply every binary active flag by its learned weight and sum the contributions.

Step 4 - Scale

Use training-sample minimum and maximum bounds to rescale raw totals to 0-100.

Step 5 - Explain

The active signal with the largest contribution becomes the customer's top risk reason. The score is a ranking index, not a probability.

Risk Tiers and Explainability

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Tiers convert a continuous ranking into operational queues with a reason attached.

Tier thresholds

- High: score of 70 or greater.
- Medium: score from 40 through 69.9.
- Low: score below 40.

Why use tiers

A retention team usually works with capacity limits. Tiers make workload planning easier, while the underlying score allows sorting within each tier.

Dominant reason

Customers receive the highest-contributing active signal as `top_risk_reason`. Examples include short tenure, electronic-check payment, or month-to-month contract.

Operational caution

Do not show the actual historical churn label to frontline users as if it were known in advance. Use risk score and reason for prioritization; keep labels for analytical validation.

Train-Test Validation

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A held-out sample provides a more honest estimate than scoring and evaluating the same records.

Split design

A deterministic customer-ID ordering assigns approximately 80% to training and every fifth customer to test. This creates 1,409 held-out customers.

Training use

Only training customers determine the overall baseline, signal churn rates, lift weights, and score scaling bounds.

Test use

Test labels are used only after scoring to calculate tier size, churn rate, precision, recall, and lift.

Why this matters

Without separation, the analysis could overstate how well the formula generalizes. A future enhancement would use time-based validation when dated snapshots become available.

Validation Results

The score meaningfully separates risk

Results below use only the untouched 20% test sample.



2.49x LIFT

High-tier churn rate divided by overall test churn rate.

50.3% CAPTURE

185 of the 368 test-sample churners appeared in High risk.

Segment Findings

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Short-tenure High-risk customers form the clearest immediate retention priority.

High-risk composition

The test sample contains 284 High-risk customers and 185 of them churned. Among these, 207 had tenure of 12 months or less as their dominant reason; 70.5% of that segment churned.

Other patterns

- High risk with electronic-check payment as the top reason: 77 customers and 50.6% churn.
- Medium risk with short tenure as top reason: 116 customers and 37.1% churn.
- Medium risk with month-to-month contract as top reason: 127 customers and 33.9% churn.

Interpretation

The strongest business story is not simply “high charges cause churn.” Risk emerges from combinations of early tenure, flexible contracts, payment method, service configuration, and charge level.

Retention Recommendations

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The score should guide experiments and service interventions, not automatic blanket discounts.

Priority 1 - Early life

Contact High-risk customers during their first year. Use onboarding check-ins, service education, installation follow-up, and issue resolution before offering price concessions.

Priority 2 - Payment journey

Test incentives for autopay or card conversion among electronic-check customers. Track adoption, churn, contact cost, and margin versus a randomized control group.

Priority 3 - Contract and service

- Offer carefully targeted contract migration to suitable month-to-month customers.
- Provide support/security education or time-limited bundle trials where those services are absent.
- Escalate fiber-service quality investigation if complaints or network data confirm an operational issue.

Measurement

Report incremental retention and contribution margin, not only contacted-customer churn. Observational association alone cannot prove an intervention will work.

Power BI Dashboard Design

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Three pages move from executive monitoring to customer action and analytical evidence.

Page 1 - Executive overview

Cards for customers, churn rate, High-risk customers, High-risk precision, and average score. Add tier distribution, test churn rate by tier, top reasons, and contract/tenure/internet slicers.

Page 2 - Retention action list

A sortable table containing customer ID, score, tier, reason, tenure, contract, monthly charges, and payment method. Default to High and Medium; use red, amber, and green conditional formatting.

Page 3 - Drivers and validation

Show signal weights, tier populations and churn rates, and a reason-by-tier matrix. Clearly label training-derived weights and test-derived performance.

Usability

Keep titles declarative, show sample definitions in tooltips, and prevent viewers from mistaking score for churn probability.

Key DAX Measures

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Simple, readable measures create consistent business definitions across dashboard pages.

Core measures

- Total Customers = `DISTINCTCOUNT(customer_risk_scores[customer_id])`
- Actual Churners filters churned to 1.
- Actual Churn Rate divides churners by total customers.
- High Risk Customers filters risk_tier to High.
- Average Risk Score averages risk_score.

Validation measure

High Risk Precision divides test-sample actual churners in the High tier by test-sample customers in the High tier. For this project it equals 65.1%.

Formatting

Format rates and precision as percentages with one decimal place. Format charges as currency. Add a custom tier sort column if Power BI otherwise orders High, Low, Medium alphabetically.

Source files

The project includes a ready-to-copy measures.dax file and dashboard_spec.md with visual-by-visual instructions.

Limitations and Production Roadmap

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The current system is a credible baseline; production requires richer behavioral data and monitoring.

Current limitations

- Single snapshot rather than dated monthly observations.
- No recent usage trends, tickets, complaints, payment failures, region, campaign cost, or outreach outcome.
- Weights represent association and the 0-100 score is not a calibrated probability.
- A deterministic random-like split is weaker than future-period validation.

Production enhancements

- Create customer-month snapshots and predict churn in a defined future window.
- Add recency, frequency, usage slope, payment failures, service incidents, and ticket sentiment.
- Compare SQL score with regularized logistic regression and tree models.
- Calibrate probabilities; tune thresholds to team capacity and intervention economics.
- Monitor drift, precision, recall, lift, calibration, fairness, and campaign incrementality.

Resume and Interview Positioning

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Describe the project accurately: an interpretable SQL risk-scoring model with held-out validation.

Resume bullets

- Built a SQL-based churn early-warning system for 7,043 telecom customers, engineering eight explainable account and service signals into a 0-100 risk score.
- Implemented an 80/20 train-test framework; the High-risk tier achieved 65.1% observed churn versus 26.1% overall, delivering 2.49x lift and capturing 50.3% of churners in 20.2% of customers.
- Produced Power BI-ready action lists, churn-driver tables, validation outputs, DAX measures, and targeted retention recommendations.

Interview structure

Explain the business decision, data limitations, feature choices, training-derived weights, test validation, quantified result, recommended experiment, and next production improvement.

Terminology

Do not call 65.1% “accuracy.” It is High-tier observed churn rate, equivalent to precision for that flagged group. Do not describe this baseline as machine learning.

Conclusion and Next Steps

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The project demonstrates end-to-end analytical thinking: modeling, validation, action, communication, and honest limitations.

What was accomplished

A reproducible SQLite pipeline converts raw customer data into explainable scores and Power BI-ready outputs. Held-out validation shows that the High tier concentrates churn at 2.49 times the baseline.

Immediate next steps

- Build the three-page Power BI report using the supplied CSV and DAX files.
- Add screenshots to the project README and publish the project repository.
- Practice a two-minute explanation using the numbers on pages 2 and 13.
- Create an A/B-test proposal for the short-tenure High-risk segment.

Final takeaway

The most valuable aspect is not the formula alone. It is the defensible chain from available data, to transparent signal engineering, to independent validation, to a realistic retention decision.

Reference

Dataset: IBM Telco Customer Churn sample. Local project materials include SQL scripts, the reproducible build, outputs, dashboard specification, DAX measures, and detailed findings.